Improving the Realism of Capital Gains Dynamics in Tax-Simulator:

A Dynamic-Programming Extension to the Realization Function

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(extended draft incorporating optimal future realization behavior)

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1 Background and Motivation

When deciding whether to sell appreciated assets, taxpayers weigh the nontax benefits of selling (portfolio rebalancing, liquidity for consumption, etc.) with the tax costs. They choose whether to sell today or to defer realization—to some future date (at which point they'll pay less tax in PV terms) or indefinitely (they'll pay no tax if basis is stepped up at death). This is a complicated decision that depends on a taxpayer's unobserved preferences, their stock of accumulated unrealized gains, their expectations about future income shocks, mortality risk, tax policy, and so on.

When tax rates change, the tax price of selling changes, so realizations change. The responsiveness of realizations to tax rates is the realization elasticity. Studies estimate the realization elasticity as a single empirical parameter (sometimes breaking it down into short-run and long-run or "permanent" components).

In Tax-Simulator, to model how capital gains realizations change in response to a tax policy change, we currently:

1. Calculate record-level changes in marginal tax rates on gains;
2. Apply a uniform realization elasticity to each of these record-level changes in marginal tax rate to generate percent change in realized capital gains;
3. Apply the percent change to each record's baseline realized capital gains and recalculate taxes.

This "top-down" or "reduced-form" method is common across the scorekeeping community. It's downstream of data and computational constraints: in the IRS PUF, we lack a longitudinal view of taxpayers and we have no information on the stock of unrealized gains.

So while the top-down application of a realization elasticity is probably good enough for approximate ten-year cost estimates of capital gains tax changes, it has several meaningful shortcomings:

- **Intertemporal accrual-realization dynamics are crudely modeled and possibly contradictory.** For decades Jane Gravelle has argued that the permanent elasticities used by scorekeepers imply that realizations exceed accruals in the long run, which is obviously impossible. Sarin, Summers, Zwick and Zidar (2022) make a similar argument: "permanent" elasticity values used in scorekeeping settings are actually more akin to short-run elasticities, in that they are estimated on time horizons insufficiently long to observe deferred realizations.

- **It does not admit changes to tax treatment at death.** Ideally, the model's behavioral response function would explicitly model the deferral decision as taking into account the tax treatment of gains at death. The top-down use of a single elasticity excludes these dynamics by construction.
- **Responsiveness is homogeneous.** A single elasticity applies to each taxpayer regardless of income, age, etc. In reality, responsiveness should depend on these features—especially if modeling fundamental changes to the tax treatment of gains at death.
- **All realization behavior comes from the intensive margin.** This is due to the multiplicative functional form assumption. In reality some people should be responding along the sell/don't-sell margin.

In the following sections I propose a ground-up realization model for Tax-Simulator that solves these problems by:

- imputing unrealized gains and mortality rates for all records;
- explicitly modeling the decision to sell, where some gains turn over for nontax reasons and the remaining discretionary realizations respond to a policy-varying tax benefit of deferral;
- ensuring accrual-realization consistency by tracking realization changes across time at the age-cohort level; and
- calibrating to aggregate elasticities and other observed behavioral moments.

Update relative to prior draft. A previous version of this writeup defined the effective tax price of realization $P_{a,t}$ using only *deterministic* future events—forced asset turnover and mortality. That specification implicitly assumed taxpayers do not anticipate making any further *discretionary* realization decisions in the future. This is internally inconsistent: a taxpayer choosing whether to sell today should rationally expect to face the same sell-or-defer choice in every future period, with policy-dependent answers. If next year's tax rate is lower, today's holder anticipates her future-self will optimally realize then, and that anticipation enters today's deferral value. The current draft replaces the closed-form $P_{a,t}$ with an effective tax price derived from a Bellman equation, solved by backward induction over the age-year grid. The prior specification turns out to be the special case in which all future discretionary realizations are zero.

2 Basic Structure

The goal is to model how a capital gains tax reform changes realizations, then track how those realization changes affect the stock of unrealized gains over time—and how that changed stock feeds back into future realizations. The model is centered around two key objects:

1. the **unrealized-gains stock equation**, a law-of-motion state variable which tracks the evolution of the policy-driven change relative to baseline in the stock of unrealized gains for each age cohort; and
2. the **realization model**, which models taxpayers' decision to sell a fraction of unrealized gains as a function of tax policy, mortality, and *forward-looking expectations of their own future realization choices.*

Tax-Simulator is a repeated cross-section: it observes records in each year, but we have no ability to follow the same taxpayer from year t to year $t+1$. The dynamic objects in this model are therefore tracked at the age-cohort level. A cell is defined by age a and year t, and the model treats cell (a,t) survivors as flowing into cell $(a+1, t+1)$.

Each cell starts with baseline quantities:

- $G^B_{a,t}$: baseline total unrealized gains
- $R^B_{a,t}$: baseline total realized gains
- $r^B_{a,t}$: baseline realization rate, $R^B_{a,t}/G^B_{a,t}$
- $m_{a,t}$: average mortality rate (policy-invariant), weighted by unrealized gains
- $\tau_{a,t}$: average marginal tax rate on gains, weighted by unrealized gains

For counterfactual scenario S, the key state variable is the change in the stock of unrealized gains relative to baseline:

$$\Delta G_{a,t} = G^S_{a,t} - G^B_{a,t}.$$

The simulator already has a baseline path $G^B_{a,t}$. The focus of our model is therefore estimating how a reform pushes the scenario stock away from that baseline path. Scenario unrealized gains are then $G^S_{a,t} = G^B_{a,t} + \Delta G_{a,t}$.

This state variable depends on how the realization rate—taxpayers' cell-aggregated sell-vs-defer decisions—responds, moving from $r^B_{a,t}$ to $r^S_{a,t}$ based on the policy change. The function governing the realization rate is the focus of Sections 4 and 5.

3 Change in the Stock of Unrealized Gains: Law of Motion

The simulator provides a baseline path for unrealized gains, $G^B_{a,t}$. The model does not attempt to re-solve that baseline path from first principles. Instead, it tracks how a reform scenario pushes the stock away from baseline.

Survivor channel. In policy-reform scenario S, the unrealized-gain stock held by surviving taxpayers in cell (a,t) after realizations is $(1-m_{a,t})(1-r^S_{a,t})G^S_{a,t}$. The corresponding baseline survivor stock is $(1-m_{a,t})(1-r^B_{a,t})G^B_{a,t}$. Subtracting baseline from scenario and substituting $G^S_{a,t} = G^B_{a,t} + \Delta G_{a,t}$:

$$(1-m_{a,t})\big[(1-r^S_{a,t})\Delta G_{a,t} + G^B_{a,t}(r^B_{a,t} - r^S_{a,t})\big].$$

To place this origin-cell contribution in the correct next-period age cell, define an aging transition matrix $A_{a\to h,t}$. With one-year age cells and an age topcode of 80,

$$A_{a\to h,t} = \begin{cases} 1 & \text{if } a < 80 \text{ and } h = a+1 \\ 1 & \text{if } a = 80 \text{ and } h = 80 \\ 0 & \text{otherwise} \end{cases}$$

The survivor contribution to destination cell $(h, t+1)$ is therefore

$$\Delta G^{\text{surv}}_{h,t+1} = \sum_a A_{a\to h,t}(1-m_{a,t})\big[(1-r^S_{a,t})\Delta G_{a,t} + G^B_{a,t}(r^B_{a,t} - r^S_{a,t})\big].$$

Death channel. Let the treatment of decedent unrealized gains be summarized by three routing shares $\delta_{\text{stepup}} + \delta_{\text{carryover}} + \delta_{\text{realize}} = 1$:

Regime	δ_{stepup}	$\delta_{\text{carryover}}$	δ_{realize}
Step-up basis	1	0	0
Carryover basis	0	1	0
Deemed realization	0	0	1

Under carryover basis, decedent unrealized gains are routed to heirs. Let $\omega_{a\to h,t}$ be the share of carryover gains from decedents of age a assigned to heirs of age h, with $\sum_h \omega_{a\to h,t} = 1$. The death-routing contribution is

$$\Delta G^{\text{death}}_{h,t+1} = \delta_{\text{carryover}} \sum_a \omega_{a\to h,t} m_{a,t} \big(G^B_{a,t} + \Delta G_{a,t}\big).$$

Full law of motion.

$$\begin{aligned}\Delta G_{h,t+1} = {} & \sum_a A_{a\to h,t}(1-m_{a,t})\big[(1-r^S_{a,t})\Delta G_{a,t} + G^B_{a,t}(r^B_{a,t} - r^S_{a,t})\big] \\ & + \delta_{\text{carryover}} \sum_a \omega_{a\to h,t} m_{a,t} \big(G^B_{a,t} + \Delta G_{a,t}\big). \end{aligned} \tag{1}$$

A few notes for intuition: if the reform equals baseline, then $r^S_{a,t} = r^B_{a,t}$ and, under current-law step-up, $\delta_{\text{carryover}} = 0$, so ΔG remains zero. Only under carryover basis do decedent unrealized gains enter next-period ΔG. Even if treatment at death is held constant, any policy-induced change in the realization rate—i.e., $r^S_{a,t} \neq r^B_{a,t}$—will register as a propagation of unrealized gains through this system.

This equation takes the scenario realization rate $r^S_{a,t}$ as given. The next two sections specify how $r^S_{a,t}$ is determined.

4 Realization Function

The starting point is a simple deferral tradeoff. A taxpayer with one dollar of unrealized gain can realize the gain today, pay tax today, and receive whatever nontax benefit comes from selling. Or the taxpayer can continue holding the asset, preserving the option to realize later or, under step-up basis, avoid tax on the embedded gain at death.

4.1 Two-channel decomposition

The model separates realizations into two channels. Some sales happen because assets turn over for reasons that are relatively less responsive to tax: business exits, divorce-driven asset divisions, life-event home sales, liquidity needs. Other sales are discretionary timing decisions that respond to the tax wedge. Let $\lambda^T_{a,t}$ denote the inelastic turnover realization rate for cell (a,t), and let $r^D_{a,t}$ denote the discretionary realization rate. Total realizations are

$$r_{a,t} = \lambda^T_{a,t} + r^D_{a,t}.$$

The turnover channel is treated as policy-invariant. The discretionary channel is the tax-responsive channel.

4.2 Functional form for the discretionary realization rate

Let $P_{a,t}$ denote an effective tax price of realizing one dollar of gain today (defined precisely below). The discretionary realization rate has the baseline-anchored form

$$r_{a,t}^{D,S} = r_{a,t}^{D,B} \exp\bigl(-\eta(P_{a,t}^{S} - P_{a,t}^{B})\bigr), \tag{2}$$

where $r_{a,t}^{D,B} = r_{a,t}^{B} - \lambda_{a,t}^{T}$ is the part of baseline realizations not attributed to turnover, and η is a calibrated response parameter. This form is unchanged from the original draft. What changes here is the definition of P, which now incorporates optimal future realization behavior.

4.3 The holder's dynamic problem

To define $P_{a,t}$, we consider a representative holder of one dollar of unrealized gain in cell (a,t). Within a period, the holder:

1. Loses fraction $\lambda_{a,t}^{T}$ to forced turnover.
2. Chooses a discretionary realization fraction $r^D \in [0, 1-\lambda_{a,t}^{T}]$, paying tax $r^D\tau_{a,t}$ and pocketing after-tax proceeds $(1-\tau_{a,t})r^D$.
3. On the remaining unrealized stock $(1-\lambda_{a,t}^{T} - r^D)$: faces mortality probability $m_{a,t}$ (with death-state tax burden $c \cdot \tau_{a,t}$ per dollar) and survives with probability $1-m_{a,t}$, carrying the surviving stock forward.

The death-state parameter is

$$c = \begin{cases} 0 & \text{under step-up basis} \\ \theta & \text{under carryover basis} \\ 1 & \text{under deemed realization at death} \end{cases}$$

where θ captures the share of the heir's future tax liability internalized by the current holder under carryover.

The holder receives a flow nontax benefit from discretionary selling, $b_{a,t}\bigl((1-\tau_{a,t})r^D\bigr)$, where the argument is *after-tax* dollars realized. This reflects that portfolio rebalancing, liquidity needs, and consumption smoothing all use after-tax proceeds.

4.4 Bellman equation

Let $W_{a,t}$ denote the per-unit value to the holder of one dollar of unrealized gain in cell (a,t). The Bellman equation is

$$W_{a,t} = \max_{r^D \in [0,\, 1-\lambda_{a,t}^{T}]} \Bigl\{ b_{a,t}\bigl((1-\tau_{a,t})r^D\bigr) - (1-\lambda_{a,t}^{T} - r^D) m_{a,t} c \tau_{a,t} + (1-\lambda_{a,t}^{T} - r^D)(1-m_{a,t})\beta W_{a+1,t+1} \Bigr\}. \tag{3}$$

Reading the right-hand side: the first term is the flow benefit from after-tax discretionary realization. The second is the death-state tax cost on the surviving (post-discretionary-realization) stock. The third is the discounted continuation value, where the surviving fraction $(1-\lambda_{a,t}^{T} - r^D)(1-m_{a,t})$ multiplies $W_{a+1,t+1}$ because the latter is per-unit and the value scales linearly with stock under the per-unit normalization.

4.5 First-order condition and effective tax price

The first-order condition with respect to r^D yields:

$$b'_{a,t}\big((1-\tau_{a,t})r^{D*}_{a,t}\big) = \frac{(1-m_{a,t})\beta W_{a+1,t+1} - m_{a,t}c\tau_{a,t}}{1-\tau_{a,t}}. \tag{4}$$

The right-hand side is the marginal opportunity cost of selling one more after-tax dollar: continuation value forgone, net of death-tax savings, divided by $(1-\tau_{a,t})$ to convert from per-unit-of-gross-realization to per-unit-of-after-tax-proceeds units.

We define the **effective tax price** as

$$\boxed{P_{a,t} \;\equiv\; \frac{(1-m_{a,t})\beta W_{a+1,t+1} - m_{a,t}c\tau_{a,t}}{1-\tau_{a,t}} \;-\; \frac{1}{\eta}\log(1-\tau_{a,t}).} \tag{5}$$

The first term is the marginal opportunity cost from the FOC. The second term is a tax-rate adjustment that arises because the holder values nontax benefits in after-tax dollars; including it is what allows the realization function to take the clean exponential form (2) after differencing scenario from baseline.

4.6 Functional form for the nontax benefit

To close the model, specify the marginal nontax benefit as

$$b'_{a,t}(x) = \mu_{a,t} - \frac{1}{\eta}\log x, \qquad x = (1-\tau_{a,t})r^D. \tag{6}$$

The intercept $\mu_{a,t}$ captures the average strength of discretionary selling motives in the cell. We do not need to specify $\mu_{a,t}$ ex ante: it is identified from baseline data via the baseline FOC.

Substituting (6) into the FOC (4) and solving:

$$r^{D*}_{a,t} = \frac{1}{1-\tau_{a,t}}\exp\big(\eta(\mu_{a,t} - P_{a,t})\big). \tag{7}$$

Taking the ratio of scenario to baseline, the $(1-\tau)^{-1}$ pre-factor and the $\log(1-\tau)/\eta$ piece of P together cancel in a way that yields exactly (2), as desired. The intercept $\mu_{a,t}$ also drops out of this ratio.

4.7 Comparison with the prior specification

The previous draft defined the effective tax price as

$$P^{\text{prior}}_{a,t} = \tau_{a,t} - \left[\sum_{j=1}^{J}\beta^j s_{a,j}\tau_{a+j,t+j} + c\sum_{j=1}^{J}\beta^j d_{a,j}\tau_{a+j,t+j}\right],$$

where $s_{a,j}$ and $d_{a,j}$ are forward-looking turnover and death hazards built from λ^T and m alone. This specification assumes that no *discretionary* realization occurs in any future period—only forced events generate future tax flows. The current $P_{a,t}$ replaces this hazard-PV expression with a Bellman recursion in which $W_{a+1,t+1}$ embeds optimal future r^{D*} choices, and therefore reflects how forward-looking taxpayers anticipate their own future selling decisions.

If we set $r^{D*}_{a+j,t+j} = 0$ for all $j > 0$, the recursion collapses and P reduces (up to the $\log(1-\tau)/\eta$ correction term) to John's P^{prior}. The current specification is thus a strict generalization: it nests the prior model as a special case in which future discretionary behavior is shut off.

5 Backward-Induction Algorithm

The Bellman equation (3) is solved by backward induction on a finite age grid. Mortality bounds the holder's lifetime, so the recursion has a true terminal condition and does not require a fixed-point iteration.

5.1 Grid setup

Extend the simulator's age grid past the topcode of 80 to a maximum age $A_{\max}$ (e.g., 110), at which mortality is approximately one. The terminal condition is

$$W_{A_{\max}+1,\, t+1} = 0 \quad \text{for all } t,$$

reflecting that beyond the maximum age, no value remains. For ages between 80 and $A_{\max}$, primitives can be extrapolated from age-80 values, with $m_{a,t}$ approaching 1 as $a \to A_{\max}$. The topcode-of-80 representation is used only for storage in the simulator; the recursion uses the extended grid.

5.2 Two-pass structure

The algorithm has two backward-induction passes: a baseline pass that computes $P^B_{a,t}$, $W^B_{a,t}$, and $\mu_{a,t}$ at every cell, and a scenario pass that computes $P^S_{a,t}$ and $W^S_{a,t}$ at every cell. The scenario discretionary realization rate is then obtained by direct application of the closed-form realization function (2):

$$r^{D,S}_{a,t} = r^{D,B}_{a,t} \exp\bigl(-\eta(P^S_{a,t} - P^B_{a,t})\bigr).$$

There is no separate FOC-solving step in Pass 2; the closed-form realization function is the FOC, in baseline-anchored form.

The intercept $\mu_{a,t}$ in the marginal-benefit function (6) cancels when scenario is differenced from baseline, but is required *within* each pass to compute $W_{a,t}$ from the Bellman equation. We recover it from the baseline FOC at every cell:

$$\mu_{a,t} = P^B_{a,t} + \frac{1}{\eta} \log\bigl((1 - \tau^B_{a,t}) r^{D,B}_{a,t}\bigr). \tag{8}$$

This computation is folded into Pass 1, since both P^B and μ depend on $W^B_{a+1,t+1}$, which is itself recursive.

5.3 Pass 1: baseline

Algorithm 1 Baseline backward induction (recovers $\mu_{a,t}$ and $W^B_{a,t}$)

Initialize $W^B[A_{\max}+1, t+1] \leftarrow 0$ for all t

for each year t in the simulation horizon **do**

  for a from $A_{\max}$ down to $a_{\min}$ **do**

    Compute baseline effective price (using already-computed $W^B[a+1, t+1]$):

$$P^B[a,t] \leftarrow \frac{(1-m[a,t])\beta W^B[a+1,t+1] - m[a,t]\,c\,\tau^B[a,t]}{1-\tau^B[a,t]} - \frac{1}{\eta}\log(1-\tau^B[a,t])$$

    Recover cell intercept from baseline FOC:

$$\mu[a,t] \leftarrow P^B[a,t] + \frac{1}{\eta}\log\big((1-\tau^B[a,t])\,r^{D,B}[a,t]\big)$$

    Compute baseline value at this cell:

$$\begin{aligned} x^B &\leftarrow (1-\tau^B[a,t])\,r^{D,B}[a,t] \\ b^B &\leftarrow \mu[a,t]\cdot x^B - \tfrac{1}{\eta}(x^B\log x^B - x^B) \\ W^B[a,t] &\leftarrow b^B - (1-\lambda^T[a,t] - r^{D,B}[a,t])\,m[a,t]\,c\,\tau^B[a,t] \\ &\quad + (1-\lambda^T[a,t] - r^{D,B}[a,t])(1-m[a,t])\beta W^B[a+1,t+1] \end{aligned}$$

return $\mu[a,t]$ for all (a,t)

5.4 Pass 2: scenario

Algorithm 2 Scenario backward induction (computes $r^{D,S}_{a,t}$ and $W^S_{a,t}$)

Initialize $W^S[A_{\max}+1, t+1] \leftarrow 0$ for all t

for each year t in the simulation horizon **do**

  for a from $A_{\max}$ down to $a_{\min}$ **do**

    Compute scenario effective price (using already-computed $W^S[a+1, t+1]$):

$$P^S[a,t] \leftarrow \frac{(1-m[a,t])\beta W^S[a+1,t+1] - m[a,t]\, c\, \tau^S[a,t]}{1-\tau^S[a,t]} - \frac{1}{\eta}\log(1-\tau^S[a,t])$$

    Compute scenario discretionary rate via the closed-form realization function:

$$r^{D,S}[a,t] \leftarrow r^{D,B}[a,t]\exp\bigl(-\eta(P^S[a,t]-P^B[a,t])\bigr)$$

    Clip to feasible range:

$$r^{D,S}[a,t] \leftarrow \min\bigl(\max(r^{D,S}[a,t],0),\, 1-\lambda^T[a,t]\bigr)$$

    Compute scenario value at this cell (using $\mu[a,t]$ from Pass 1):

$$\begin{aligned}
x^S &\leftarrow (1-\tau^S[a,t])\, r^{D,S}[a,t] \\
b^S &\leftarrow \mu[a,t]\cdot x^S - \tfrac{1}{\eta}(x^S\log x^S - x^S) \\
W^S[a,t] &\leftarrow b^S - (1-\lambda^T[a,t]-r^{D,S}[a,t])\, m[a,t]\, c\, \tau^S[a,t] \\
&\quad + (1-\lambda^T[a,t]-r^{D,S}[a,t])(1-m[a,t])\beta W^S[a+1,t+1]
\end{aligned}$$

return $r^{D,S}[a,t]$ for all (a,t)

The scenario discretionary rate $r^{D,S}_{a,t}$ output by Algorithm 2 feeds directly into the law of motion in Section 3 as $r^S_{a,t} = \lambda^T_{a,t} + r^{D,S}_{a,t}$.

5.5 Practical notes

Computational cost. With ages from 18 to $A_{\max} = 110$ and a 30-year simulation horizon, the grid has roughly 2,800 cells per pass. Each cell requires a handful of arithmetic operations. Total cost is on the order of milliseconds per scenario.

Numerical stability. The $\log r^{D,B}$ term in $\mu_{a,t}$ blows up if any baseline cell has zero realizations. In practice, floor $r^{D,B}$ at a small positive number (e.g., 10^{-6}) before applying (8).

Calibration of η. Because P is more responsive to tax-rate changes than P^{prior} (since changes propagate through future r^{D*}), the value of η that matches a given target aggregate semi-elasticity will differ from the value calibrated under the prior specification. Calibration should be re-run using Algorithms 1–2: perturb tax rates slightly under current-law death treatment, compute the implied change in aggregate realizations, and choose η to reproduce the target.

Sanity check. Setting $r_{a,t}^{D,B} = 0$ everywhere as a diagnostic should produce a P that collapses to the prior hazard-PV expression (up to the $\log(1-\tau)$ correction). This is a useful unit test of the implementation.

6 Model Properties

Returning to the motivations at the top of this writeup, a useful model of realizations should make three predictions: step-up reduces the level of realizations for a given tax-rate regime; step-up raises the realization elasticity (the revenue-maximizing rate is higher absent step-up); and realizations deferred due to a tax-rate increase today, if not held until death, show up as realizations in some future year. The mechanisms are summarized below.

Prediction	Mechanism
Step-up lowers realization level	Death-state value in $W_{a+1,t+1}$ raises the value of holding under step-up; raising c lowers W and raises P
Step-up raises elasticity	Shutting the death escape valve makes deferral more rate-sensitive; the bracket multiplying η in the elasticity formula rises with c
Deferred realizations resurface later	Forgone realizations stay in ΔG via the law of motion, raising future G^B and feeding back into future realization decisions

The dynamic-programming formulation strengthens the third channel materially: deferred realizations resurface not just through the stock law of motion, but also through forward-looking holders who anticipate future favorable tax conditions and actively defer toward them. A full re-derivation of the comparative statics in this section under the new P is left to a follow-up note.

7 Calibration

Model parameters are either observed in baseline tax data, assigned judgmentally, or calibrated to match aggregate moments.

Object	Role	Source
$G^B_{a,t}$	Baseline unrealized gains	Baseline PUF projections (via SCF imputations)
$R^B_{a,t}$	Baseline realized gains	Baseline PUF projections
$r^B_{a,t}$	Baseline realization rate	Derived: $R^B_{a,t}/G^B_{a,t}$
$m_{a,t}$	Mortality rate	Baseline PUF projections (via SSA life table imputations)
$\tau_{a,t}$	Marginal tax rate on gains	Tax-Simulator calculation
$\lambda^T_{a,t}$	Turnover realization rate	Asset-class assumptions or data
$\mu_{a,t}$	Cell-level discretionary intercept	Recovered in Pass 1 from baseline FOC
η	Tax-price response parameter	Calibrated to aggregate elasticity target
β	Discount factor	Assumption (e.g., 0.96)
c, θ	Death-treatment parameters	Regime-determined; θ is an assumption
$\omega_{a\to h,t}$	Heir age routing	Implied estate and inheritance flows from SCF

The response parameter η is calibrated so that the model matches a target aggregate semi-elasticity under current law. Let $\varepsilon^{\text{target}}$ denote the target. Under current-law step-up, choose η such that

$$\frac{\partial \log \sum_{a,t} R^S_{a,t}}{\partial \tau} = \varepsilon^{\text{target}}.$$

This is solved numerically: perturb capital gains tax rates slightly under current-law death treatment, run Algorithms 1–2, compute the implied change in aggregate realizations, and choose η so the model reproduces the target.

The discount factor β controls how much future tax payments matter for today's realization decision. A starting value around 0.96 corresponds to a four percent annual discount rate, but this should be varied in sensitivity analysis.